

The Kodamai Container Program: A Complete Technical Record

Part II — Composition: Welds, Distributive Laws, the Obstruction Cluster

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Abstract

Part II records the composition theory of the container program. Composing two directed containers (= small categories = systems) that share an interface is a *Zappa–Szép product* $C \bowtie D$, equivalently a distributive law; existence is governed by two checkable conditions; when the second fails the obstruction is a cohomology class. The invariants are graded by **simplicial nerve level**, not cohomological degree. Every result carries its trust grade verbatim and its registry node. A critical 2026-10-09 correction (Rick UID 211) on the meaning of the H^2 weld class is preserved throughout.

Grade legend. [peer-reviewed] > [proved] > [lean-verified]/[computed] > [peer-claimed]/[in-progress]
Grades are copied verbatim and never inflated.

Thesis of Part II. Composing two directed containers that share an interface is a Zappa–Szép product $C \bowtie D$, equivalently a distributive law. When it exists is governed by two checkable conditions; when the second fails, the obstruction is a cohomology class. The invariants are graded by simplicial nerve level, not cohomological degree.

1 Orchestration = Zappa–Szép weld

Precis. Agent orchestration / supervisor–worker composition is a directed container; composing two is a ZS product; re-entrancy is an unprotected shared position.

Theorem 1.1 (Orchestration is a ZS weld, [proved]). *Orchestration is a directed container; composition = the ZS product $C \bowtie D$; re-entrancy is a failed guard. Node `orchestration-zs`.*

Theorem 1.2 (Pairwise ZS criterion, [proved]). *A matched pair extends to a global ZS product iff (L) freeness and (G) global closure both hold. Node `pairwise-zs`.*

Theorem 1.3 (ZS associativity, [lean-verified]). *$ZS1-ZS4 \iff$ associativity (`ZappaSzep.lean`; see Part V).*

2 The H^2 weld obstruction — with the 2026-10-09 correction

Precis. Condition (G) is governed by a class $[\omega] \in H^2$.

Flag (CRITICAL CORRECTION, 2026-10-09, Rick UID 211). The cocycle *condition* IS associativity; the *class* $[\omega]$ measures the **SPLITTING** of an already-associative weld, **NOT** non-associativity. The honest non-associativity separator is a pointwise N_3 magma condition (a condition, not a class). State $[\omega]$ only as a splitting obstruction throughout.

Theorem 2.1 ($(G) \iff [\omega] = 0$, [proved]). $(G) \iff [\omega] = 0$ with $[\omega] \in H^2(\text{Sk}_C; \mathcal{D})$; the class is the splitting obstruction of an associative weld. Connections *g-obstruction-is-baues-wirsching*, *h2-homes-for-zs-obstruction*.

Theorem 2.2 (Groupoid case, [computed]). For a connected groupoid $(= K(\Gamma, 1))$ the ZS defect is a Schreier factor set and merge $\iff [\omega] = 0$; moreover $cd \leq 1 \implies [\omega] = 0$ (free groupoids always merge). Node *groupoid-zs-obstruction*.

Theorem 2.3 (Two $[\omega]$ sites, [proved]). The two $[\omega]$ sites — handoff $[\omega_h]$ and stabiliser $[\omega_{st}]$, both $\cong \mathbb{F}_2$ — are irreducibly distinct. Node *two-omega-sites*.

Flag (DEMOTED 2026-10-09). The weld- ω -as-non-associativity gloss (and the re-entrancy-stub reading) went [proved]→[computed]; state $[\omega]$ only as a splitting obstruction.

3 Holonomy and the Zappa–Szép bridge

Precis. Composing update monads sharing a state set is a ZS product that welds state/isotropy holonomy; the emergent-holonomy meeting invariant is cohomological.

Theorem 3.1 (Holonomy composition ZS bridge, [peer-reviewed]). Composing two update monads sharing $S = a$ ZS product on the position-threading monoid, welding holonomy; the aligned-abelian class $[\omega] \in H^2(B; A)$ obstructs splitting. Rick-refereed. Node *holonomy-composition-zs-bridge*.

Theorem 3.2 (Holonomy-triviality collapse, [proved]). The holonomy-triviality collapse engine: *liftings* $\cong \text{Cat}$ completeness. Node *holonomy-triviality*.

Theorem 3.3 (Emergent-holonomy Ext tower, [proved]). The emergent-holonomy meeting count is $\text{Ext}_{kG}^n(k[G/A], k[G/B]) \cong \oplus H^n$ — the Ext tower. Node *emergent-holonomy-is-ext-tower*.

4 The nerve-level grading of compositional invariants

Precis. Grade an invariant of a small category by the **simplicial nerve level** it reads, not by cohomological degree. Regrade lemma: cohomological degree $n = \text{nerve level} - 1$, so composition (composable pairs N_2) is **degree ONE**, not two. This retires the shipped “composition is a degree-two datum” slogan.

Theorem 4.1 (Nerve-level tower, [proved]). Node *composition-nerve-level*. (Rung 1) N_1 reflexive-graph / U -blind invariants (out-degree, H^0) sit **STRICTLY** below (rung 2) N_2 composition invariants (H^1 , δ -translations, inventory $[\theta_R]$); (rung 3) N_3 associativity. **Thm 3:** a category is 2-coskeletal, so the truncation tower collapses at level 2 (no third truncation rung). **Thm 4:** the genuine strict $N_2 \subsetneq N_3$ lives in the **ASSEMBLY**/weld problem, where associativity is an N_3 predicate not implied by N_2 data.

Flag (Precision, Rick N1, 2026-10-09). The shape invariant $U(p)$ is strictly coarser than $\text{tr}_1 N$ (both $[1]$ and $B(\mathbb{Z}/2) + \text{pt}$ give $U = y^2 + y$) — do NOT call U “the nerve-level-1 invariant.” **Terminology (N3):** “nerve level” must be disambiguated (truncation level vs. cocycle-condition level) throughout.

5 Skew-brace / solvability obstruction arc

Precis. The deeper algebraic home of the ZS obstruction: skew braces, the residue-dependent solvability obstruction o_ν , and Ferri prunability.

Theorem 5.1 (Solvability obstruction o_ν , [proved]). $o_\nu: H_+^2 \rightarrow C^3/A(Z_\circ^2)$ is a group homomorphism with $\text{im } \varphi^+ = \ker o_\nu$; nontrivial $\implies$ torsor. Node *solvability-obstruction-o-nu*. (Several finite slices are [lean-verified] — Part V.)

Theorem 5.2 (Ferri prunability, [computed]). A Ferri “completely prunable” quiver skew brace = vanishing of the ZS global obstruction; prunability = degree-1 descent/subfunctor, NOT Schreier; with a complete-prunability no-go. Node *quiver-skew-brace-zs*.

6 Effects / coeffects and the composition of (co)monads

Precis. One Set-monad M gives two container feeds (effect monad T_M , coeffect comonad G_M); they always entwine, the arrow face needs non-branching, and state lives on $\otimes$ not $\triangleleft$.

Theorem 6.1 (Monad-to-comonad transfer, [proved]; [lean-verified]). A monad on Set transfers to a comonad $G(S, P) = (S, M \circ P)$. Node *monad-comonad-transfer* (*MonadComonadTransfer.lean* / *DualTransfer.lean*).

Theorem 6.2 (Effect/coeffect entwining, [proved]). The bialgebra face holds for all M ; the arrow/biKleisli face holds iff non-branching; the branching obstruction = Atkey’s index. Nodes *monad-comonad-entwining*, *effect-coeffect-arrows*.

Theorem 6.3 (State object, [proved]). The state object $\Delta S =$ the codiscrete category, giving the store/costate comonad; *state via* $\otimes$, *not* $\triangleleft$. Node *state-object-delta*.

Theorem 6.4 (Workers grading, [proved]; [lean-verified]). The Workers $(\text{Set}, \times)$ -graded category, its type hierarchy, and the retract into the BHM $\triangleright$ -grading. Nodes *workers-type-hierarchy*, *workers-retract-of-bhm-grading* (*WorkersRetract.lean*).

7 Game comonads (adjacent)

Precis. The finite-model-theory game comonads factor through Poly, connecting the composition story to logic/complexity.

Theorem 7.1 (Game comonads, [computed]). The Ehrenfeucht–Fraïssé E_k , pebbling P_k , and modal M_k comonads factor through Poly / are polynomial comonoids. Node *game-comonad-poly*.